MILITARY SPECIFICATION

DISPLAY UNIT, SIGHT, HELICOPTER HEAD UP SUBSYSTEM: M76

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Departmet of Defense.

1. SCOPE.

1.1 Scope. This specification contains quality assurance provisions and preparation for delivery for the Display Unit, Sight, Helicopter, Head Up Subsystem: M76.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-E-6051	Electromagnetic Compatibility Requirements, Systems
MIL-I-8500	Interchangeability and Replaceability of Components Parts for Aircraft and Missiles
MIL-Q-9858	Quality Control Requirements
MIL-F-13926	Fire Control Materials: General Specification Governing the Manufacture and Inspection of
MIL-O-13830	Optical Component for Fire Control Instrument, General Specification Governing Assembly and Inspection of

FSC 1270

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-L-25467	Lighting, Integral Aircraft Instrument, General Specification for
MIL-M-38510	Microcircuits, General Specification for
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication and Radar for Aerospace and Control Systems, Procedure for
MIL-S-46844	Solder Bath, Soldering of Printed Wiring Assemblies, Automatic Machine Type
MIL-W-63150	Weapons and Support Material, Standard Quality Assurance Provisions for
MIL-S-70442	Sight, Helicopter, Head-Up Subsystem: M76

STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods

2.1.2 Other Government documents, drawings and publications. The following other other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

11780001	Display Unit, Sight, Helicopter Head-Up Subsystem
11780731	HUD Block Diagram

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein:

American Society for Testing and Materials (ASTM).

ASTM - D3951 - Standard Practice for Commercial Packaging.

3. REQUIREMENTS.

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications and standards.

3.2 Parts. All parts shall comply with applicable drawings, specifications, and standards.

3.3 Head up display (HUD). The HUD shall comply with all requirements specified on Drawing (Dwg) 11780001, all associated drawings, and with all requirements specified in applicable specifications and standards. The HUD covered by this specification shall be configured for use on the AH-1S series helicopter and is part of MIL-S-70442 (Sight, Helicopter, Head-Up Subsystem: M76). The HUD shall display a visual presentation of flight and fire control data to the pilot during day and night missions utilizing a four inch optics aperature. The data shall be collimated and superimposed on the pilot's image of the real world by means of a partially reflective optical image combiner. The displayed data shall be as commanded from the Signal Processor (SP) Dwg. 11780002. For a functional block diagram that shows the general signal flow for the HUD unit see Dwg. 11780731.

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3.3.1 Major components list. Major components of the HUD are as follows:

Nomenclature	Drawing
CCA, X-Deflection	11780024
CCA, Y-Deflection	11780060
CCA, Bright-Up Ampl	11780011
CRT Assy	11780036
Chassis Assy, HUD	11780064

3.4 Characteristics.

3.4.1 Electrical.

3.4.1.1 Electrical signal interface. The HUD signal interface with the AH-1 series helicopter shall be as defined in Appendix A.

3.4.1.2 Electrical power.

3.4.1.2.1 Steady state. The electrical service shall be supplied to the HUD from the Signal Processor (SP). Edge Lit Panel power shall be supplied from aircraft. The following powers shall be supplied:

Signal Name	Nominal Voltage (±2%)	Maximum Continuous Current	% Regulations 10% to Full Load
Filament PWR	+9V	0.55A	6%
HVPS PWR	+26.5V	0.75A	3%

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Signal Name	Nominal Voltage (±2%)	Maximum Continuous Current	% Regulations 10% to Full Load
Deflection Boost PWR	.40V	0.10A	3%
Deflection PWR	+16V	0.4A	3%
Deflection PWR	+16V	0.4A	3%
Z-Amp PWR	+85V	0.04A	3%
Z-Amp PWR	-85V	0.003A	10%
Edge Lit Panel	0 to 28V	TBD	-

3.4.1.2.2 Electrical power interruptions. The HUD shall return to specified operation within 5 seconds after momentary interruption as defined in MIL-STD-704. HUD shall return to specified performance within 15 seconds after long term interruption.

3.4.1.3 Visual interface.

3.4.1.3.1 HUD visual interface. The HUD shall present a minimum visual obstruction and safety hazard to the pilot, when mounted on the pilot's instrument panel.

3.4.1.3.2 Night vision goggles interface. The HUD symbols and manual controls shall be clearly visible when viewed with the AN/PVS-5 night vision goggles. The spectral emission and/or the adjustable brightness range of the HUD symbols and manual controls shall be such that adjustment of the instrument panel lighting controls and the HUD symbol brightness control accomplishes compatibility with the goggles. Spectral filtering shall be allowed

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in meeting this requirement. A dimming discrete shall be sent to the HUD to permit switching normal night level illumination to a nominally low level illumination compatible with night goggles.

3.4.1.4 Mechanical alignment. The HUD shall allow mounting to a boresighted HUD mounting tray (11780246) without additional boresighting.

3.4.2 Performance.

3.4.2.1 Operation. The HUD is used by the pilot to display flight and weaponry information. The pilot observes these data items from his normal head position. The pilot shall have control of display operation with controls located on the HUD, and shall control displayed data by the use of externally generated command signals.

3.4.2.2 HUD optical performance.

3.4.2.2.1 HUD field-of-view (FOV).

3.4.2.2.1.1 Instantaneous FOV. The HUD shall have a minimum apparent instantaneous binocular rectangular rield of view of $\pm$96 milliradians (mr) in azimuth and $\pm$113 mr in elevation with respect to the aircraft datum line (ADL) when viewed binocularly from the design-eye position using an eye separation of 2.5 inches.

3.4.2.2.1.2 Total FOV. The HUD shall have a total apparent FOV of at least 20 degrees (349 mr) minimum with the optics axis being 1.5 degrees from the ADL (0.0 degrees). The total FOV shall be observable from the alternate eye position defined to be 8.0 inches forward of the design eye position and within the head motion box shown in Figure 1.

3.4.2.2 Collimination and parallax.

3.4.2.2.1 Collimination. The HUD symbology shall be colliminated to appear to be focused at infinity.

3.4.2.2.2.2 Parallax. The maximum parallax (accept/reject criteria) permitted from the design eye position shall be as follows:

a. Vertical disparity: $\leq$1 mr
b. Horizontal convergent display: $\leq$2 mr
c. Horizontal divergent disparity: $\leq$1 mr

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3.4.2.2.3 Optical transmissibility. The combining glass shall have a minimum optical transmissibility of 75 percent required, over the FOV of the HUD.

3.4.2.2.4 Combining glass deviation error. Each combining glass shall not cause a radial angular displacement of the real world of more than 0.25 mr. This requirement shall apply for all eye positions that are required to obtain the total FOV.

3.4.2.2.5 Geometric distortion and instability. The geometric distortion, jitter and instability of the individual symbols over all electrical and environmental ranges of operation shall be consistent with the HUD accuracy requirements.

3.4.2.2.6 Optical seal. The optical module shall be suitably sealed to assure that all entering air will pass through a passive desiccator. The desiccator shall, when properly regenerated, prevent internal fogging of the optics during variations of temperature, altitude and humidity. The mounting shall allow inspection of the desiccant condition with the pilot's display unit installed in the helicopter. Removal and replacement shall be accomplished without disassembly or removal of the HUD from the helicopter.

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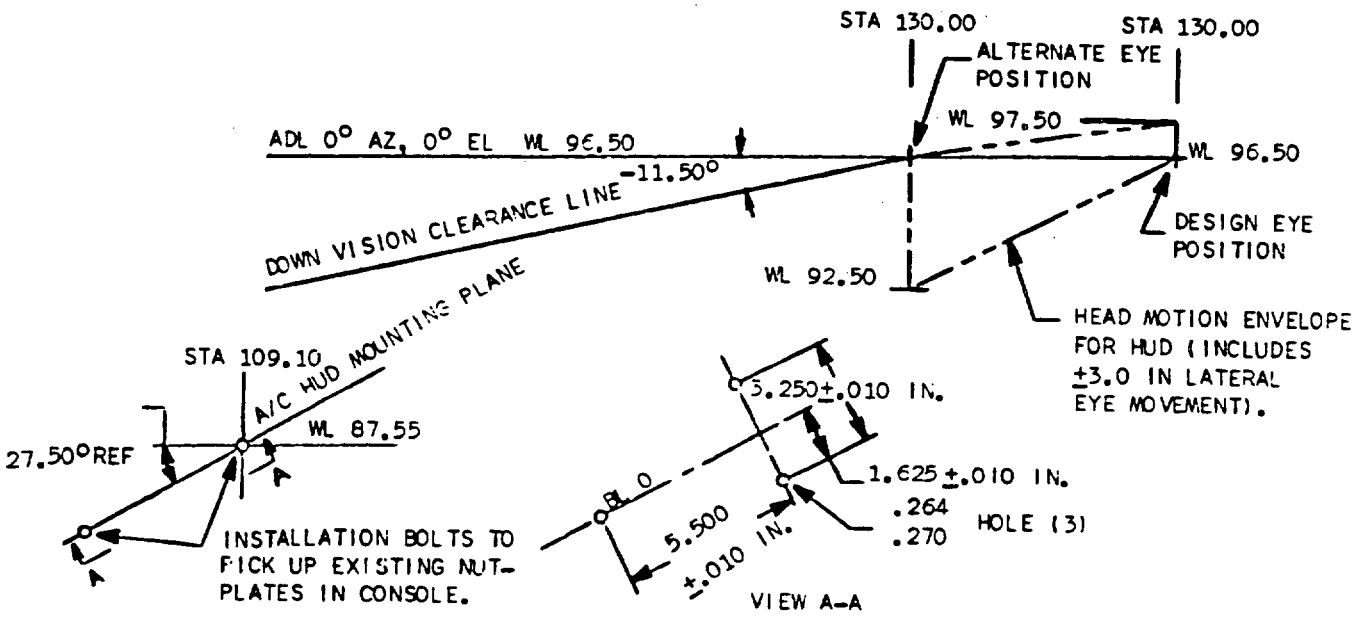


FIGURE 1. HUD OPTICAL INSTALLATION CONSTRAINTS

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3.4.2.3 Data characteristics.

3.4.2.3.1 Data definition. Data signal characteristics supplied to the HUD are as stated in Table I.

3.4.2.3.2 Symbol and reticle line width. The observed line width of the HUD reticles or symbol shall be 1.0 mr with a tolerance of ±0.5 mr at the 50 apercent line intensity point for a display brightness setting in the range of 10 foot-lambert to 1000 foot-lamberts.

3.4.2.3.2.1 Displayed data color. The color of the displayed data shall be consistent with the human factor and visual interface requirements of this specification.

3.4.2.3.3 Brightness.

3.4.2.3.3.1 Displayed data brightness. The line brightness of all displayed data shall be such that the images shall be clearly visible against a 10,000 foot-lambert background birghtness, and in the presence of direct sunlight penetration in the cockpit. The minimum contrast ratio (C) under the conditions, shall be 1.15. The contrast ratio (C) is defined as:

$$C = \frac{B_2 + B_1}{B_1}$$

Where: B_2 is the apparent brightness of the symbology as viewed through the combiner.

B_1 is the apparent brightness of the background as viewed through the combiner.

The projected image shall be clearly defined with no visible breaks or irregularities when viewed from a normal operating position in a darkened area or against a bright sky.

3.4.2.3.3.2 Brightness control range. The displayed data brightness shall be adjustable by controls and/or filtering to accommodate and ambient brightness range of 1×10^{-4} ($1X10^{-5}$ desired to 10,000 foot-lamberts in two ranges if necessary. For ambient brightness below 1.0 foot-lambert, AN/PVS-5 goggles may be worn by the pilot. Symbol brightness shall vary non-linearly with adjustment of the manual brightness control on the HUD display unit, such that the control sensitivity at a given contrast is approximately constant over the ambient brightness range.

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Table I. Data Characteristics

Signal	Parameters
Deflection X + Deflection X -	Input impedance 3.9K ohm, differential, analog signal (horizontal axis). ± 5 Volt signal each line, ± 10 Volts differential. ± 10V diff = ±10.5° deflection (183.259 mr).
Deflection Y + Deflection Y -	Input impedance 3.9K ohm, differential, analog signal (elevation axis). ± 5 Volt signal each line, ± 10 Volts differential. ± 10V diff = ±10.5° deflection (183.259 mr).
Bright-up +	Input impedance 680 ohm + 5 V dc = Bright-up, O V dc = Cut-off
Bright-up -	Input impedance 680 ohm O V dc = Bright-up, + 5 V dc = Cut-off

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3.4.2.3.3.3 Brightness uniformity. For a given brightness setting, the average line brightness of symbols shall not vary more than 25 percent over the FOV of the HUD. Brightness variation is defined as follows:

$$B \text{ variation} = \frac{B \text{ max} - B \text{ min}}{B \text{ max} + B \text{ min}}$$

Where:

B max = maximum symbol line brightness

B min = minimum symbol line brightness

3.4.2.3.3.4 Automatic brightness control (ABC). A light sensor and control system which automatically adjusts the brightness of the HUD and symbology in response to the outside ambient light level shall be incorporated in the HUD. A switch on the HUD front panel permits the pilot to enable or defeat the automatic control. The ABC shall operate within the brightness range of the symbols and for an ambient range of 100 to 6000 foot-lamberts.

3.4.2.4 Accuracy. Within the environmental conditions specified herein, the HUD shall not exceed 1.7 mr error, one signma, in each axis of azimuth and elevation for the FCS retical.

3.4.2.5 Power control modes. The HUD shall have a toggle switch for the use of power control mounted on its front panel.

a. OFF. This shall be the down position. OFF shall cause the HUDS to have no power on its circuits.

b. STANDBY. This shall be the middle position. STANDBY shall invoke the SP to supply power to all circuits and blank the Cathode Ray Tube (CRT).

c. ON. This shall be the up position. ON shall cause the SP to render the HUDS system totally operational.

3.4.2.5.1 Operational modes. The HUD shall have a toggle switch for the use of mode selection mounted on its front panel. The switch shall have three positions as listed below:

a. NORMAL. This shall be the up position. NORMAL shall cause the SP to suppy appropriate signals to the HUD for display of normal mode symbology.

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b. TEST. This shall be the middle position. TEST shall command the SP to supply appropriate signals to the HUD for displaying the test pattern.

c. STAD. This shall be the down position. STAD shall induce the SP to supply appropriate signals to the HUD for displaying the Statiametric Reticle (SMR).

3.4.2.5.2 Automatic brightness control switch. The Automatic Brightness Control (ABC) circuitry shall be controlled by a two position toggle switch on the HUD front pannel. The up position shall initiate the ABC. The down position shall defeat ABC.

3.4.2.5.3 Brightness control. This shall be a potentiometer adjustment for adjusting display birghtness in an out of the ABC mode. The potentiometer shall be located on the left side of the HUD as viewed by the pilot.

3.4.2.5.4 Direct/indirect fire mode. The HUD shall incorporate a two position toggle switch which shall be a part of the HUD controls for use in helicopter fire control circuits.

3.4.2.5.5 Night filter. The night filter is for use in conjunction with the brightness control potentiometer for night flying and use with AN/PVS-5 night vision goggles. The night filter control shall be a knob, located behind the Brightness Control knob on the left hand side of the HUD as viewed by the pilot.

3.4.2.6 Display charactersitics.

3.4.2.6.1 Ambient light reflections. The HUD optics shall be designed to avoid ambient light reflections from the combining glass and HUD optics. The HUD optics shall be positioned with consideration given to minimizing sunlight reflected into the pilot's eyes.

3.4.2.6.2 Secondary images. Brightness levels of any secondary images shall be less than 3 percent of the primary image brightness over the total FOV and for head locations anywhere within the head motion box indicated in Figure 2.

3.4.2.6.3 Reflector/combiner plate. The combiner glass shall conform to MIL-O-13830.

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3.4.2.6.4 Night symbol color. To accomodate night operation without the AN/PVS-5 night vision goggles, it is required that the symbols displayed on the HUD shall be red in color in accordance with MIL-L-25467.

3.4.2.7 Built-in-test equipment (BITE).

3.4.2.7.1 Fault detection/location. The HUD shall have a Built-In-Test (BIT) capability of checking its operational status. BUT shall obviate the need for any external test equipment for the performance of preflight or in-flight testing of the HUD unit.

3.4.2.7.1.1 Test accuracy. The accuracy of BIT stimuli and measurements shall be designed to preclude false pass/fail indications. When the equipment is operated within the environmental and service conditions specified herein, less than 2.0 percent of all "fail" indications shall occur when no fault exists.

3.4.2.7.1.2 Fault detection and isolation capability. BIT shall provide the crew with a minimum of 95 percent confidence level that no failure which would result in a loss of function or mode in flight would occur.

3.4.2.7.1.3 BIT reliability. BIT shall be designed so that any failure of the BIT circuitry will not inherently cause failure or degradation of the HUD's primary performance as defined herein.

3.4.2.7.1.4 Malfunction indication. An equipment status indicator shall be provided for the HUD. The equipment status indictor shall be a go/no-go device and shall be such as to hold the last result if power is interrupted or removed, and shall be on the surface of the HUD.

3.4.2.7.1.5 BIT output. The BIT circuitry shall also provide a HUD fail output signal to the SP in order to initiate a HUD's no-go signal and hold the SP BIT indicator from false tripping.

3.4.3 Physical characteristics.

3.4.3.1 Dimensions. Dimensional requirements applicable to envelope size, mounting and mating shall be in accordance with Drawing 11780001.

3.4.3.2 Weight. The weight of the HUD shall be 14.25 pounds maximum.

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3.4.3.3 Panel lighting. The instrument shall be provided with integral 5-volt red lighting conforming to MIL-L-25467, except as follows:

a. Coating. Reflection reducing coating shall be in accordance with MIL-C-14806.

b. Reduced voltage operation. The lighting shall operate satisfactorily at reduced voltage. The relative brightness and color of the display shall be uniform and all markings shall be legible when the lighting is operated at 2.70 ± 0.02 Vdc and viewed from a distance of 28 to 30 inches.

3.4.3.4 Control panels. All switches, controls, and displays shall be easily accessible and operable for left-hand operation with standard issue winter gloves.

3.4.4 Preconditioning (burn-in). Each HUD procured under this specification shall be subjected to a burn-in test as defined in Appendix B.

3.4.5 Environmental conditions (first article testing only). The HUD shall operate and meet the performance requirements of 3.4.2 under any of the environmental conditions, or a reasonable combination of these conditions, as specified herein.

3.4.5.1 Natural environment. The HUD shall operate and withstand storage in the environments specified herein.

3.4.5.1.1 Altitude. Operating: 0-20,000 feet; nonoperating and storage: 0-50,000 feet.

3.4.5.1.2 Temperature. Operating nonoperating, and storage: -50° to +71°C.

3.4.5.1.3 Humidity. Operation while being subjected to MIL-STD-810, Method 507.1, Procedure II. In Step 6, measurements shall be taken during the last five hours of the last test cycle.

3.4.5.1.4 Sand and dust. Operation after being subjected to MIL-STD-810, Method 510.1, Procedure I, except that the air velocity in Steps 1, 2, and 3 shall be 200 ± 100 fpm. Step 3 may proceed immediately after stabilization in Step 2, if desired.

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3.4.5.1.5 Salt atmosphere. Operation after being subjected to MIL-STD-810, Method 509.1. Operation of the HUD shall not be required and failure criteria shall be limited to corrosion of finished and metals only. Such corrosion shall be defined as visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface.

3.4.5.1.6 Fungus. No evidence of viable fungus on any surfaces after being subjected to MIL-STD-810, Method 508.1.

3.4.5.1.7 Explosive atmosphere. Operation after being subjected to MIL-STD-810, Method 511.1, Procedure I, except test to be limited to a maximum altitude of 20,000 feet.

3.4.5.1.8 Rain. Operation after being subjected to MIL-STD-810, Method 506.1, Procedure I. During each thirty minute rain period, the wind shall be maintained for fifteen minutes, commencing five minutes after the start of the period as specified in the procedures. The HUD shall be operated during the last ten minutes of the thirty minute period.

3.4.5.2 Service environment. The HUD shall withstand the service environment as specified in MIL-E-5400 for Class 2 equipment except as noted in Paragraphs 3.4.7.1.1 and 3.4.7.1.2.

3.4.5.3 Induced environment.

3.4.5.3.1 Vibration. The HUD shall not sustain damage or subsequently fail to provide specified performance when subjected to the vibration environment defined by Method 514.2, Procedure I, Part I, Table 514.2-III of MIL-STD-810; except that the frequency shall be limited to 500 Hz (reference Figure 18).

3.4.5.3.2 Shock. The HUD shall meet the shock requirements of MIL-STD-810, Method 516.2, Procedure I, Figure 516.2-2 and Method 516.2, Procedure III.

3.4.5.3.2.1 Bench handling. The HUD shall withstand shock environments encountered in maintenance and servicing.

3.4.5.3.3 Acceleration. The HUD shall operate without degration when exposed to acceleration levels up to 2g.

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3.4.6 Transportability. The HUD shall withstand vibration environments encountered during transporting as specified in MIL-STD-801, Method 514.2, Procedure X.

3.4.7 Soldering. All soldering shall be done in accordance with MIL-S-45743 or MIL-S-46844.

3.4.8 Electromagnetic radiation (first article testing only).

3.4.8.1 Electromagnetic compatibility. Each HUD component as installed in the helicopter, shall not cause undesirable response, malfunction, or degraded performance to any other equipment/ subsystem installed in or associated with the helicopter; nor shall any component malfunction, exhibit degraded performance or undesirable response beyond the tolerances specified in MIL-E-6051 when other equipment on board the helicopter are singly or collectively operated and performing their intended functions.

3.4.8.2 Electromagnetic interference. The HUD shall meet the requirements of MIL-STD-461 equipment when tested to the following requirements: CE01, CE03, CE05, CS01, CS02, CS06, RE02, RS01, RS03, RS03.1, and RS02.

3.4.8.3 Circuit isolation. Signal and power potentials shall be electrically isolated from each other and the equipment chassis.

3.4.9 Interchangeability (first article only). All components having the same manufacturer's part number shall be dimensionally and functionally interchangeable in accordance with MIL-I-8500. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part number and changes thereto. Interchangeability must be maintained within the detail parts to the level of replaceable items.

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article by the contractor shall be as specified in the contract.

3.6 Workmanship. Workmanship shall be in accordance with electronic and mechanical weapon-build practices to ensure safety, proper and dependable operation, and service life. Particular attention shall be given to assure thoroughness of assembly, alignment of parts, electrical connections, cleanliness, tightness of screws and bolts, marking of parts, application of protective finish, and freedom from burrs and rough or sharp edges.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classed as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-E-5400.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three HUDs selected at random by the Government representative from the first fifteen production HUD.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in Table III.

4.4.2 Failure. Failure of any HUD to meet the requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any HUD to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

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TABLE II. First article tests.

Item	Characteristic	Requirement	Test Procedure
1	Natural environment	3.4.5.1	4.6.3.1
2	Altitude	3.4.5.1.1	4.6.3.2
3	Temperature	3.4.5.1.2	4.6.3.3
4	Humidity	3.4.5.1.3	4.6.3.4
5	Sand and dust	3.4.5.1.4	4.6.3.5
6	Salt atmosphere	3.4.5.1.5	4.6.3.6
7	Fungus	3.4.5.1.6	4.6.3.7
8	Explosive atmosphere	3.4.5.1.7	4.6.3.8
9	Rain	3.4.5.1.8	4.6.3.9
10	Interchangeability	3.4.11	
11	Proof test	3.6	
12	Service environment	3.4.5.2	4.6.3.10
13	Vibration	3.4.5.3.1	4.6.3.10.1
14	Shock	3.4.5.3.2	4.6.3.10.2
15	Bench handling	3.4.5.2.1	4.6.3.11.2.1
16	Acceleration	3.4.5.3.3	4.6.3.11.2
17	Transportability	3.4.6	4.6.3.11.3
18	Electromagnetic radiation	3.4.8	4.6.3.12

TABLE III. Performance tests.

Item	Characteristic	Requirement	Test Procedure
101	Electrical signal interface	3.4.1.1	Appendix A
102	Electrical power	3.4.1.2	4.6.2.1
103	Operation	3.4.2.1	4.6.2.2
104	HUD optical performance	3.4.2.2	4.6.2.3
105	Data characteristics	3.4.2.3	4.6.2.4
106	Accuracy	3.4.2.4	4.6.2.5
107	Power control modes	3.4.2.5	4.6.2.6
108	Display characteristics	3.4.2.6	4.6.2.7
109	Built-in-test equipment (BITE)	3.4.2.7	4.6.2.8
110	Preconditioning (burn-in)	3.4.4	Appendix B

TABLE IV. General tests.

Item	Characteristic	Requirement	Test Procedure
301	Fabrication and General Specifications	3.1, 3.2, & 3.3	Applicable drawings visual, and MIL-E-5400, Visual

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4.5 Quality conformance inspection.

4.5.1 Performance inspection.

4.5.1.1 Sample and tests. Each HUD in every lot shall be subjected to the tests in Table IV and shall be examined visually for completeness, improper assembly, and evidence of poor workmanship.

4.5.1.2 Acceptance. Where any HUD fails to meet any specified requirement, the defective HUD shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.2 General inspection.

4.5.2.1 Sample and tests. One HUD, as a control sample, shall be selected at random by the Government representative from each 100 HUDs produced and shall be subjected to the tests in Table V.

4.5.2.2 Acceptance. Where any HUD fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subjected to the provisions of MIL-STD-105.

4.6 Methods of inspection.

4.6.1 Test equipment. The test equipment in Table V shall be used where specified to perform the required tests.

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TABLE V. Test equipment.

Item/Nomenclature	Type		Mfr. & Model or Part No
	Standard	Peculiar	
1. Digital Voltmeter	X		HE 3440A with Range Unit
2. Oscilloscope	X		Tektronix 547 with W Preamp
3. HUDS Test Set(M142)		X	11780600
4. Power Test Panel		X	Ohio Semitronics PTP 707
5. Optical Boresight Set including:			(TBD)
Boresight Bench			
Teodolite			Zeiss 70-24-15
Autocolimating Eye Piece			Zeiss 70-44-10
6. High Efficiency Photometric Telescope	X		Gamma Scientific 2020-31 SMR x/Microscope Eyepiece
7. Comptroller	X		Gamma Scientific DR1A
8. X/Y Plotter	X		HP 7045A-002
9. Ammeter	X		Commercial
10. Cable (W2)		X	11780654
11. Cable (W3)		X	11780656
12. Cable (W4)		X	11780657
13. Cable (W5)		X	11780658
14. Flashlight	X		Commercial

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4.6.2 Performance tests.

4.6.2.1 Electrical power. Use items 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 and 13 of Table V to determine compliance with 3.4.1.2.

4.6.2.2 Operation. Use items 3, 4, 10, 11, 12 and 13 of Table V to determine compliance with 3.4.2.1.

4.6.2.3 HUD optical performance. Use items 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 and 13 of Table V to determine compliance with 3.4.2.2.

4.6.2.4 Data characteristics. Use items 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 and 13 of Table V to determine compliance with 3.4.2.3.

4.6.2.5 Accuracy. Use items 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 and 13 of Table V to determine compliance with 3.4.2.4.

4.6.2.6 Power control modes. Use items 3, 4, 5, 10, 11, 12 and 13 of Table V to determine compliance with 3.4.2.5.

4.6.2.7 Display characteristics. Use items 3, 4, 5, 6, 7. 8, 9, 10, 11, 12 and 13 of Table V to determine compliance with 3.4.2.6.

4.6.2.8 Built-in-test equipment (BITE). Use items 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 and 13 of Table V to determine compliance with 3.4.2.7.

4.6.3 Environmental tests.

4.6.3.1 Natural environment. Subject the system to the following environments.

4.6.3.2 Altitude. The system shall be subjected to the tests of Method 504.1 of MIL-STD-810 for Category 4 equipment. The system, nonoperating, shall be subjected to Method 500.1, Procedure I, with pressures simulating air transport at 50,000 feet altitude at a pressure of 2.85 inches of mercury (hg) for 1 hour. At the conclusion of this test, subject the system to the performance tests of Table IV.

4.6.3.3 Temperature. The system shall be subjected to test Method 501.1, Procedure II, of MIL-STD-810, except Step 4 temperature shall be 71°C. The system shall then be subjected to the test of

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Method 502.1, Procedure I, except maximum low temperature shall be -54°C. At the conclusion of this test, subject the system to the performance tests of Table IV.

4.6.3.4 Humidity. The system shall be subjected to test Method 507.1, Procedure II, of MIL-STD-810. In Step 6, measurements shall be taken during the last five hours of the last test cycle. At the conclusion of this test, subject the system to the performance tests of Table IV.

4.6.3.5. Sand and dust. The system shall be subjected to test Method 510.1, Procedure I, of MIL-STD-810, except that the air velocity in Steps 1, 2, and 3 shall be 200 ± 100 fpm. Step 3 may proceed immediately after stabilization in Step 2, if desired. At the conclusion of this test, subject the system to the performance tests of Table IV.

4.6.3.6 Salt atmosphere. The system shall be subject to the test of Method 509.1 of MIL-STD-810. Operation of the subsystem shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface.

4.6.3.7 Fungus. The system shall be subjected to the test of Method 508.1, Procedure I, of MIL-STD-810. Gasket sealed or other non-hermetic sealed test items shall be opened during test exposure and all internal and external surfaces shall be sprayed with spore suspension. Operation of the equipment is not required, but examination of the concentration after exposure period shall reveal no evidence of viable fungus on any surfaces.

4.6.3.9 Explosive atmosphere. The system shall be subjected to test Method 511.1, Procedure I, of MIL-STD-810, except test shall be limited to a maximum altitude of 20,000 feet. At the conclusion of this test, subject the system to the performance tests of Table IV.

4.6.3.9 Rain. The system shall be subjected to test Method 506.1, Procedure I, of MIL-STD-810. During each thirty minute rain period, the wind shall be maintained for 15 minutes, commencing 5 minutes after the start of the period as specified in the procedures. The system shall be operated during the last 10 minutes of the thirty minute period. At the conclusion of this test, subject the system to the performance tests of Table IV.

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4.6.3.10 Service environment.

4.6.3.10.1 Vibration. The system shall be subjected to the vibration test of MIL-E-5400, Figure 2 (Sheet 1 of 2), Curve III, except that the maximum double amplitude shall be 0.20 inch. At the conclusion of this test, subject the system to the performance tests of Table IV.

4.6.3.10.2 Shock. The system shall be subjected to the shock test of Paragraph 3.2.2.4.6.1 of MIL-E-5400. The tests of Paragraph 3.2.2.4.6.2 of MIL-E-5400 for mounting hardware crash safety shall apply only to the units mounted inside the cockpit. At the conclusion of this test, subject the system to the performance test of Table IV.

4.6.3.11 Induced environment.

4.6.3.11.1 Vibration. The system shall be subjected to test Method 514.2, Procedure I, Part I, Table 514.2-III, of MIL-STD-810, except that the frequency shall be limited to 500 cycles (reference Fig. 2). At the conclusion of this test, subject the system to the performance test of Table IV.

4.6.3.11.2 Shock. The system shall be subjected to:

a. Method 516.2 of MIL-STD-810, Procedure I, Figure 516.2-2 for flight vehicle equipment. The test "g" value shall be within ± 10 percent when measured with a 0.2 to 250 Hz filter. The pulse test level and duration are defined in Paragraph 3.2.24.6.1 of MIL-E-5400.

b. Method 516.2, Procedure III for mounting base crash safety. The pulse test level and duration are defined in Paragraph 3.2.24.6.2 of MIL-E-5400.

At the conclusion of this test, subject the system to the performance test of Table IV.

4.6.3.11.2.1 Bench handling. The system shall be subjected to Method 516.2, Procedure V, of MIL-STD-810. At the conclusion of this test, subject the system to the performance tests of Table IV.

4.6.3.11.3 Transportability. The system shall be subject to the test of Method 514.2, Procedure X, of MIL-STD-810. At the conclusion of this test, subject the system to the performance tests of Table IV.

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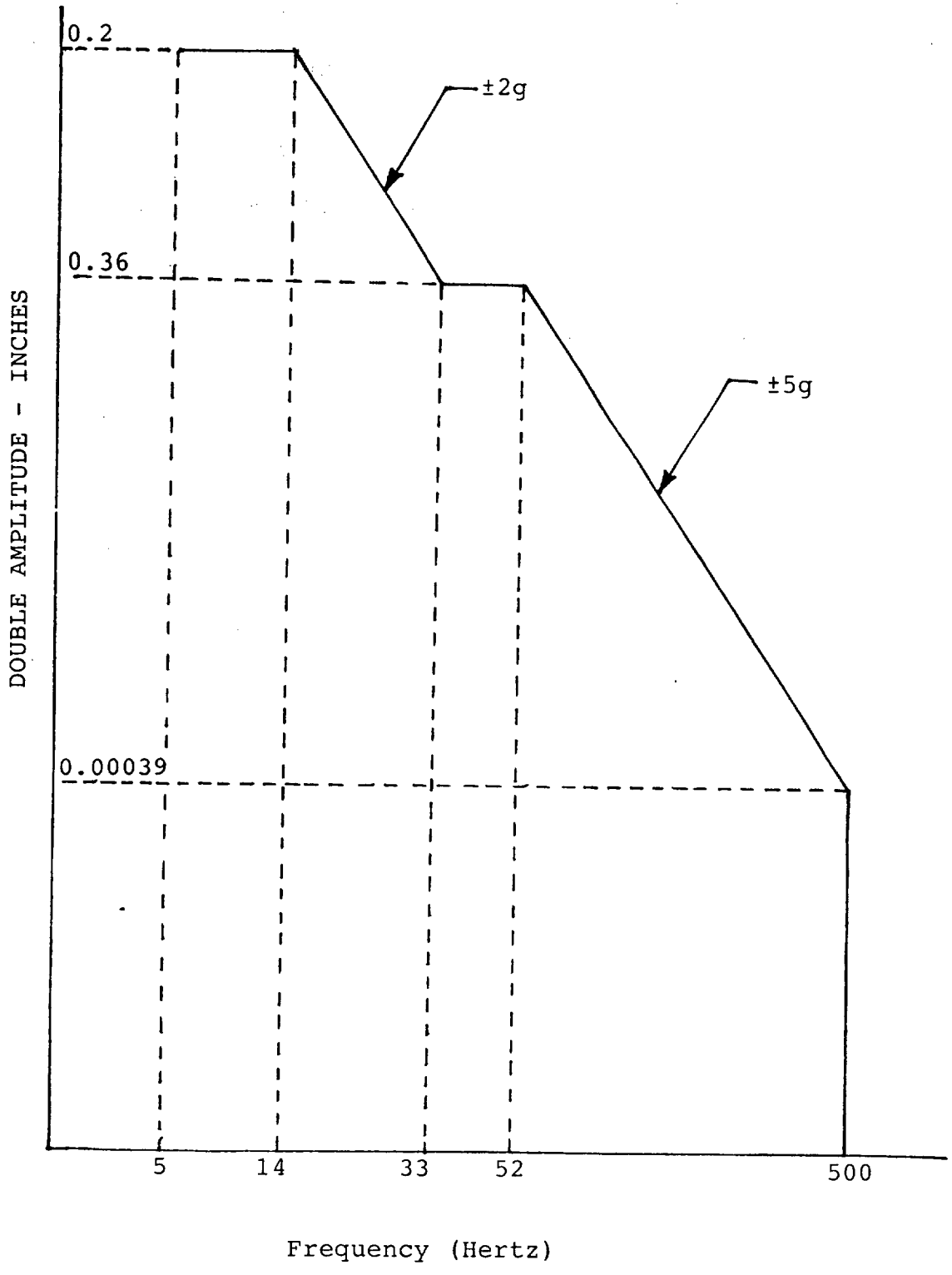


FIGURE 2. Vibration Amplitude for First Article Test.

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5. PACKAGING.

5.1 Preservation, packing, packaging, and marking. The Display Unit, Sight, Helicopter Head-Up Subsystem: M76; Dwg 11780001 shall be preserved, packaged, packed and marked in accordance with the requirements of the preservation and packaging data sheet for P11780001 at the level of protection specified in the contract (see 6.2b).

5.2 Commercial packaging shall be in accordance with ASTM-D3951.

6. NOTES.

6.1 Intended use. This equipment is intended for use in the AH-1S (modernized) helicopters.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Applicable packaging Data Sheet (see 5.1).

c. Selection of applicable levels of preservation, packaging and packing.

d. Applicable stock number.

e. First article inspection requirements.

6.3 Previous revisions of MIL-D-70443 were for the US Army Armament Research and Development Center internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1270-A050)

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APPENDIX A

HUDS SIGNAL INTERFACE

REQUIREMENTS

Detail characteristics. The HUD communicates with the Fire Control Computer (FCC), the TOW Missile System (TMS), the AN/APM 209 radar altimeter, Airborne Laser Tracker (ALT), and the Laser Range Finder (LRF). Data transmission from the FCC and from the LRF is by means of dedicated serial-data busses. Signals from the TMS and the radar altimeter are in the form of parallel discrete and analog signals.

HUD-FCC interface. The FCC transmits data to HUD by means of dedicated serial digital bus. The line consists of a twisted, shielded wire pair which carries the data and its logical complement. The wiring configuration is shown in Figure 3. A timing diagram illustrating typical signal timing is shown in Figure 4.

Bus characteristics. The data line-pair between the FCC and the HUD consists of a twisted shielded wire pair with a characteristic impedance of 71 ohms, ±10 percent. The shield on this pair is grounded in accordance with good EMI suppression practices. Data is transmitted and received by differential 3-state logic transmitters and receivers. The signal levels used shall be TTL compatible.

Data format. The data is sent asynchronously at a normal 9600 bit/second rate. Each data byte forms a part of an 11-bit transmission period consisting of a start bit, 8-bit data byte, stop bit, and parity bit, as shown in the timing diagrams of Figure 4. Transmitted parity, including parity bit, is odd. Two such bytes sent sequentially make up a 16-bit word.

Message format. The information sent in the data words are in the sequence shown in Table I. The FCC sends all six words, two bytes per word, each data transmission.

HUD-LRF interface. The LRF transmits data to the HUDS on a serial digital data bus like that for the FCC. The wiring configuration is as shown in Figure 3. In the event of FCC failure, data transmission is to be received directly from the LRF with signal timing as shown in Figure 5.

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HUD-TMS interface. The HUDS replaces the Pilot's Steering Indicator (PSI) in the TMS and so receives all the signal information which the PSI would normally receive.

HUD-radar altitude interface. The HUDS receives low and high limit discretes directly from the radar altimeter. Radar altitude comes to the HUDS as a digital work in the data from the FCC. In case of FCC malfunction, radar altitude information may be absent. The HUDS continue to generate high and low warning symbols normally in response to these discretes in that case.

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TABLE II. Data Word Structure.

Parameter	Displayed Resolution	Parameter Resolution	Parameter Range	Number of Bits Required	Byte Positions in Data Transfer Sequence
Range Smoothed	10m	1m	0.2 to 9.99 km	14 Bits	1 and 2
Radar Altitude	5 ft	1 ft	0 to 5000 ft	13 Bits	3 and 4
Magnetic Heading	1°	1°	0 to 359°	9 Bits	5 and 6
Torque	1%	1%	0 to 145%	10 Bits	7 and 8
FC Reticle AZ	0.3582 mr	0.1 mr	±157 mr	11 Bits + Sign (Extended sign two's complement)	9 and 10
FC Reticle EL	0.3582 mr	0.1 mr	+131 mr -184 mr	11 Bits + Sign (Extended sign two's complement)	11 and 12
TSU AZ	0.3582 mr	0.1 mr	±110°	15 Bits + Sign (Extended sign two's complement)	13 and 14
TSU EL	0.3582 mr	0.1 mr	+30° -60°	14 Bits + Sign (Extended sign two's complement)	15 and 16

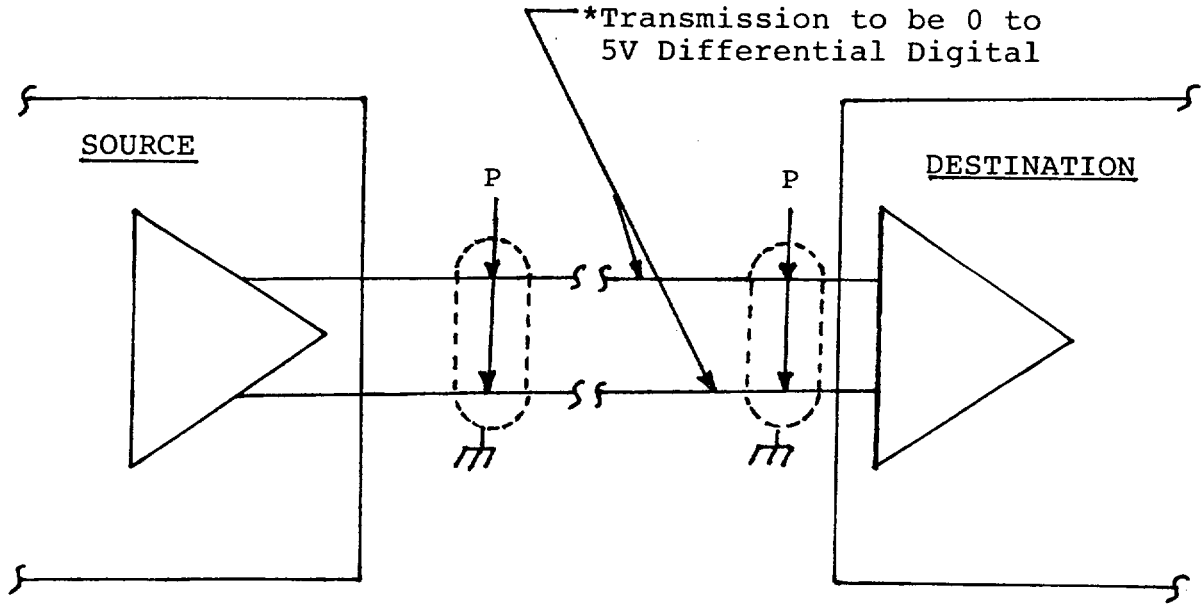
DATA BLOCK TRANSFER

byte 1	byte 2	byte 3	byte 4	byte 5	byte 6	byte 7	byte 8	byte 9	byte 10	byte 11	byte 12	byte 13	byte 14	byte 15	byte 16

Start of Data Block

End of Data Block

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OUTPUT CAPABILITY

Fairchild 9614 or Equivalent

Baud Rate = 9600

INPUT CAPABILITY

Fairchild 9615 or Equivalent

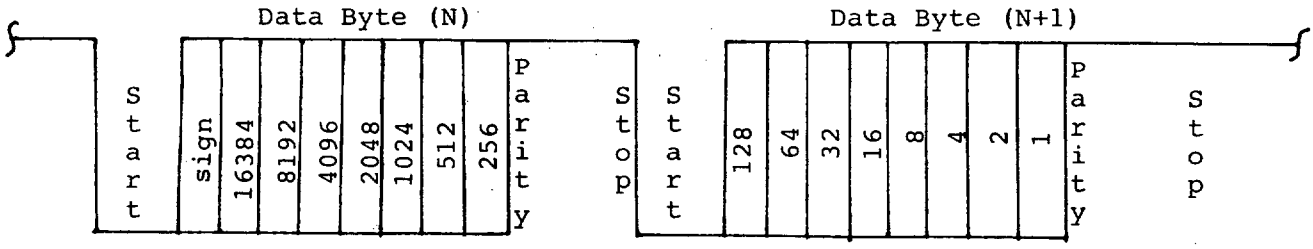
Baud Rate = 9600

*

	LINES	
	TRUE	FALSE
LOGIC "1"	+2.8 To +5.0 V	-0.5 To +0.8 V
LOGIC "0"	-0.5 To +0.8 V	+2.8 To +5.0 V

FIGURE 3. Interface Bus.

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Where: N = 1, 3, 5, 7, 9, 11
N+1 = 2, 4, 6, 8, 10, 12

Note: All Data is to be 16 BIT 2's Complement Binary Data Scaled from the LSB (1) as defined in Parameter Resolution Column of Table I.

Each Byte is 11 Bits.

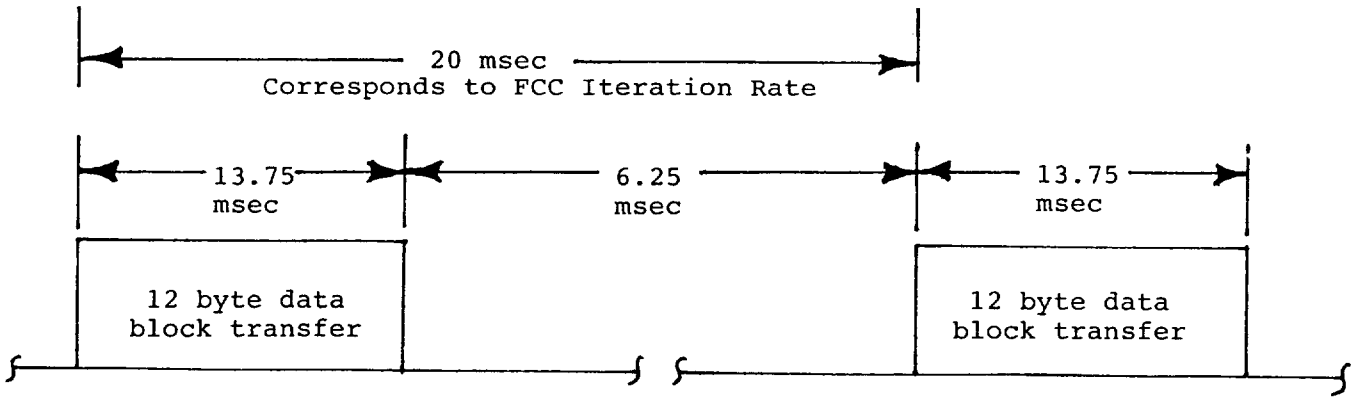
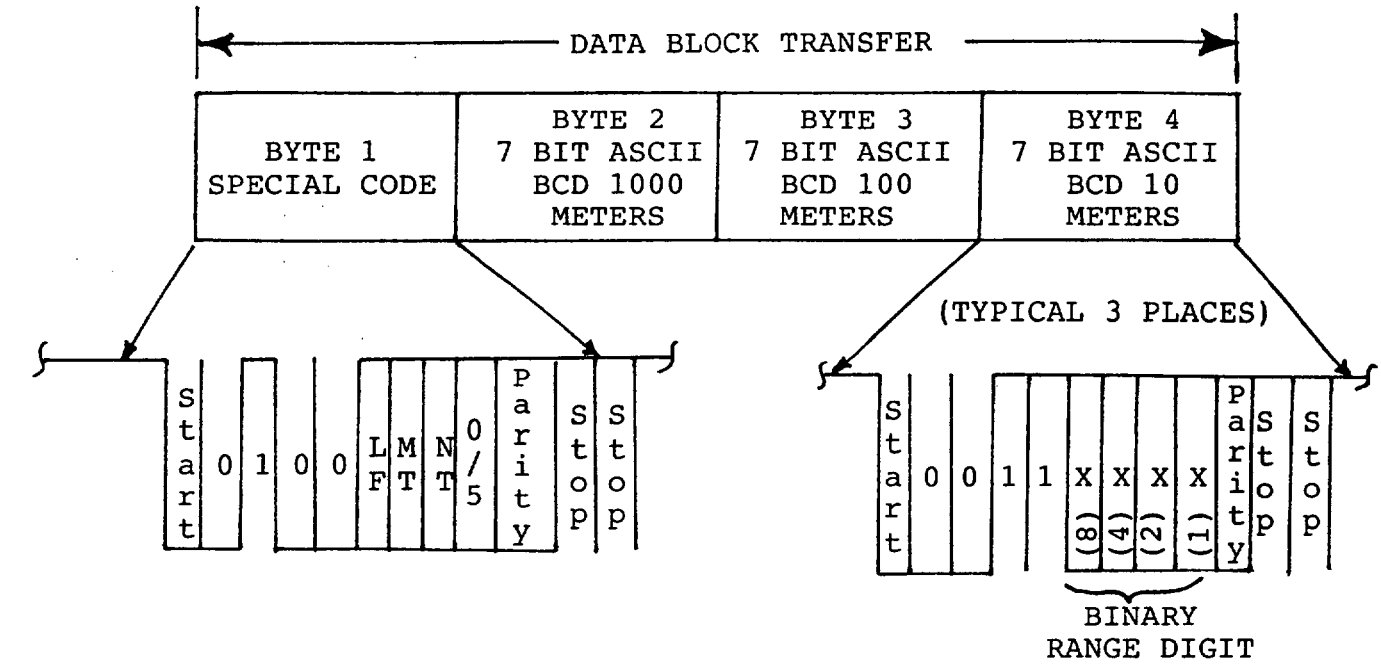
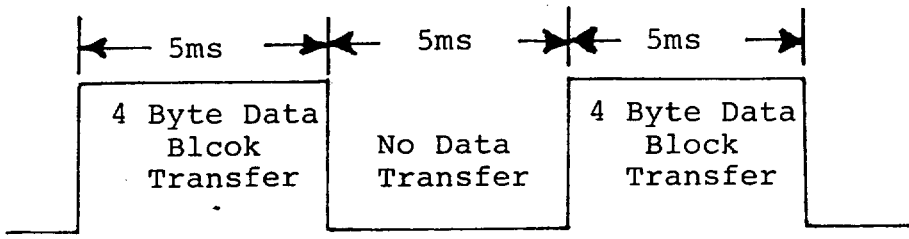


FIGURE 4. Data Transmission from FCC to HUD.

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DATA TRANSFER TIMING



SPECIAL CODE DEFINED

LF = Laser Fired When True
MT = Multi-Target When True
NT = No Target Return When True
0.5 = 5 Meter Incr.

ODD TRUE PARITY

FIGURE 5. Laser Range Finder Data Format.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 20

NOMENCLATURE	FCC GOOD
TYPE	28 VDC
CONDUCTOR TYPE AND GAUGE	SINGLE UNSHIELDED
ORIGIN	FIRE CONTROL COMPUTER
DESTINATION	SEE REMARKS
DESTINATION IMPEDANCE	SEE REMARKS
SOURCE	FCC, RELAY
SOURCE IMPEDANCE	N/A
MAX CURRENT	0.25 AMPERE TOTAL
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	+28 VDC (FCC GOOD)
FALSE STATE VOLTAGE	OPEN CIRCUIT (FCC FAIL)

REMARKS DESTINATIONS AND DESTINATION IMPEDANCES ARE:

1. IFCU, RELAY COIL, 186 OHMS OR GREATER, WITH SURFACE DEVICE TO AVOID TURN-OFF TRANSIENTS.

2. PILOT'S CAUTION PANEL, RELAY COIL, 550 OHMS OR GREATER, WITH SUITABLE DEVICE TO AVOID TURN-OFF TRANSIENTS.

3. HUDS, $\geqq$6.6 K OHMS.

4. ROCKET MANAGEMENT SUBSYSTEM, $\geqq$2K OHMS.

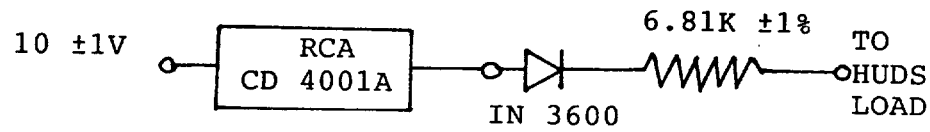
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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 32

NOMENCLATURE	DESCENT COMMAND
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	TOW/SCA
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geq$20 K OHMS
SOURCE	TOW/SCA
SOURCE IMPEDANCE	7K OHMS (SEE REMARKS)
MAX CURRENT	1 MA
MAX VOLTAGE	11V
TRUE STATE VOLTAGE	+10 $\pm$ 1V (DESCEND)
FALSE STATE VOLTAGE	OPEN CIRCUIT

REMARKS

10 VDC CAUSES HUD TO DISPLAY DESCEND COMMAND SYMBOL WHEN IN TOW MODE OF FIRE.



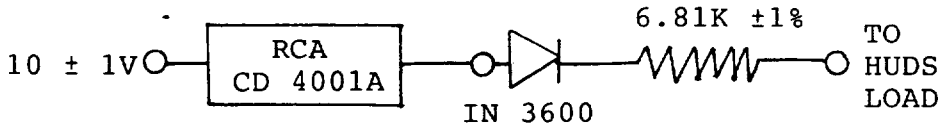
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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 33

NOMENCLATURE	ASCEND COMMAND
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	TOW/SCA
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geq$20 K OHMS
SOURCE	TOW/SCA
SOURCE IMPEDANCE	7K OHMS (SEE REMARKS)
MAX CURRENT	1 MA
MAX VOLTAGE	11V
TRUE STATE VOLTAGE	+10 $\pm$ 1V (ASCEND)
FALSE STATE VOLTAGE	OPEN CIRCUIT

REMARKS

10 VDC CAUSES HUD TO DISPLAY ASCEND COMMAND SYMBOL WHEN IN TOW MODE OF FIRE.



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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 34

NOMENCLATURE	ATTACK MODE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	AIRCRAFT 28 VDC
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geq$6.6K OHMS
SOURCE	TOW/SCA
SOURCE IMPEDANCE	$<$100 OHMS
MAX CURRENT	5 MA
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	28 VDC(ATTACK)
FALSE STATE VOLTAGE	OPEN CIRCUIT

REMARKS

28 VDC CAUSED HUD TO DISPLAY TOW MODE PRELAUNCH ATTACK SYMBOLS.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 35

NOMENCLATURE	READY MODE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	AIRCRAFT 28 VDC
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geq$6.6K OHMS
SOURCE	TOW/SCA
SOURCE IMPEDANCE	$\leq$100 OHMS
MAX CURRENT	5 MA
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	28 VDC (READY)
FALSE STATE VOLTAGE	OPEN CIRCUIT

REMARKS

28 VDC CAUSED HUD TO DISPLAY TOW MODE PRELAUNCH ATTACK SYMBOLS.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 36

NOMENCLATURE	FIRE/MANEUVER MODE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	AIRCRAFT 28 VDC
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geq$6.6K OHMS
SOURCE	TOW/SCA
SOURCE IMPEDANCE	$<$100 OHMS
MAX CURRENT	5 MA
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	28 VDC (MANEUVER)
FALSE STATE VOLTAGE	OPEN CIRCUIT

REMARKS

AFTER THE TOW MISSILE HAS BEEN FIRED, A 28 VDC DISCRETE CAUSES THE HUD TO DISPLAY TOW MODE POSTLAUNCH SYMBOL.

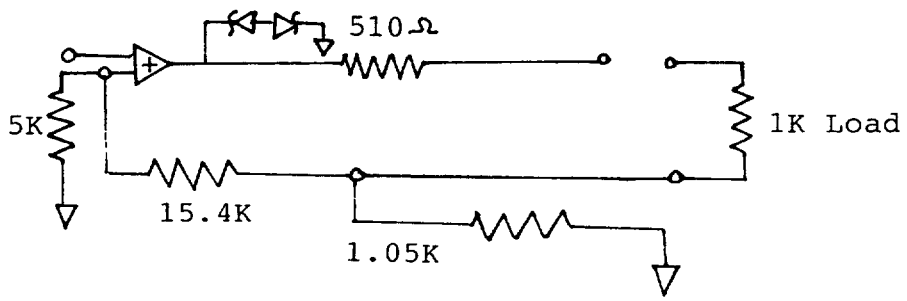
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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 37

NOMENCLATURE	TSU AZ ANGLE
TYPE	VARIABLE DC (CURRENT DRIVEN)
CONDUCTOR TYPE AND GAUGE	TWISTED PAIR SHIELDED
ORIGIN	TOW/SCA
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	1K OHMS
SOURCE	TOW/SCA
SOURCE IMPEDANCE	(CURRENT SOURCE)
VOLTAGE RANGE	N/A
SCALE FACTOR	SEE REMARKS
PARAMETER RANGE	SEE REMARKS
PARAMETER ACCURACY	SEE REMARKS
PHASING	+ CURRENT IS LOS TO THE RIGHT
RMS NULL	N/A
PHASE SHIFT	N/A
SENSITIVITY	N/A
MAX CURRENT	± 2.2 ma (ALSO SEE REMARKS)
MAX VOLTAGE	± 2.2 VOLTS (ALSO SEE REMARKS)
MAX SLEW RATE	SEE REMARKS

REMARKS

1. PRELAUNCH AND NON-TOW MODE CUEING SIGNAL CHARACTERISTICS: ±0.4 MA/DEGREE ±4% OVER ±3.75 DEGREES AT ±250 MR/SEC MAX.
2. POSTLAUNCH-SIGNAL CHARACTERISTICS (TOW MODE ONLY): ±0.0136 MA/DEGREE ±5% OVER ±110 DEGREES AT ±1.7 RAD/SEC MAX.
3.



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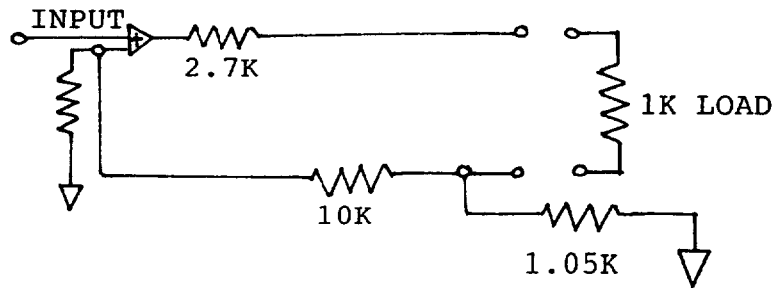
DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 38

NOMENCLATURE	TSU EL ANGLE
TYPE	VARIABLE DC (CURRENT DRIVEN)
CONDUCTOR TYPE AND GAUGE	TWISTED PAIR SHIELDED
ORIGIN	TOW/SCA
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	1K OHMS
SOURCE	TOW/SCA
SOURCE IMPEDANCE	(CURRENT SOURCE)
VOLTAGE RANGE	N/A
SCALE FACTOR	SEE REMARKS
PARAMETER RANGE	SEE REMARKS
PARAMETER ACCURACY	SEE REMARKS
PHASING	+ CURRENT IS LOS UP
RMS NULL	N/A
PHASE SHIFT	N/A
SENSITIVITY	N/A
MAX CURRENT	± 2.2 ma (ALSO SEE REMARKS)
MAX VOLTAGE	± 2.2 VOLTS (ALSO SEE REMARKS)
MAX SLEW RATE	SEE REMARKS

REMARKS

THE NORMAL OPERATING RANGE FOR THIS SIGNAL IS:

1. PRELAUNCH AND NON-TOW MODE CUEING SIGNAL CHARACTERISTICS: ±0.1 MA/DEGREE ±4% OVER ±7.5 to -15 DEGREES AT ±250 MR/SEC MAX.
2. POSTLAUNCH SIGNAL CHARACTERISTICS (TOW MODE ONLY): ±0.027 MA/DEGREE ±10% OVER ±30 to -60 DEGREES AT ±1.7 RAD/SEC MAX.
3.



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ANALOG SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 39

NOMENCLATURE	AZ COARSE POINTER
TYPE	VARIABLE DC (VOLTAGE DRIVEN)
CONDUCTOR TYPE AND GAUGE	TWISTED PAIR SHIELDED
ORIGIN	TOW/SCA
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	1K OHMS $\pm$10%
SOURCE	TOW/SCA
SOURCE IMPEDANCE	$\geq$5.76K OHMS
VOLTAGE RANGE	$\pm$1.5 VDC
SCALE FACTOR	.0136 v/DEG
PARAMETER RANGE	$\pm 110^{\circ}$
PARAMETER ACCURACY	$\pm$10% OF ACTUAL
PHASING	+ VOLTAGE IS LOS TO THE RIGHT
RMS NULL	N/A
PHASE SHIFT	N/A
SENSITIVITY	N/A
MAX CURRENT	$\pm$2.2 ma
MAX VOLTAGE	$\pm$2.2 VDC

REMARKS

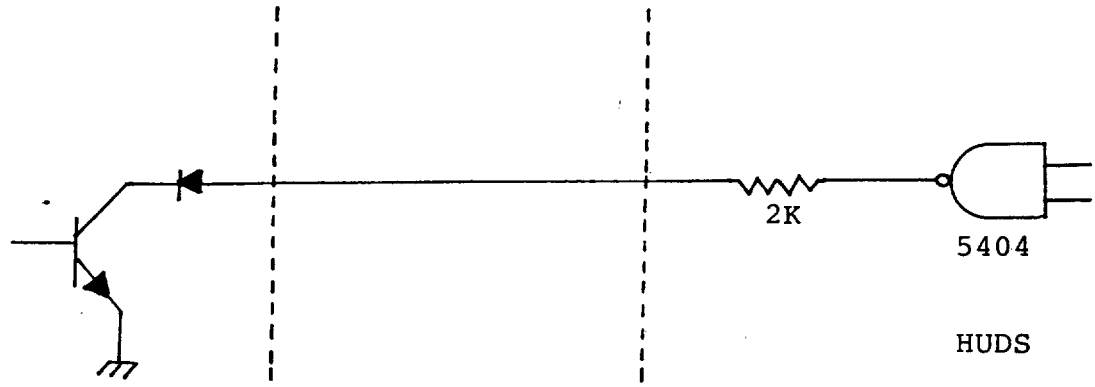
This symbol is used to position the TSU AZ LOS coarse symbol on the HUD when the gunner's LOS symbol is being used on the HUD.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 40

NOMENCLATURE	LOW ALTITUDE LIMIT
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	(SEE REMARKS)
SOURCE	RADAR ALTIMETER - APN-209
SOURCE IMPEDANCE	N/A
MAX CURRENT	10 ma
MAX VOLTAGE	5.5 VDC
TRUE STATE VOLTAGE	TRANSISTOR SWITCH CLOSED (ALTITUDE BELOW LIMIT)
FALSE STATE VOLTAGE	TRANSISTOR SWITCH OPEN (ALTITUDE ABOVE LIMIT)

REMARKS



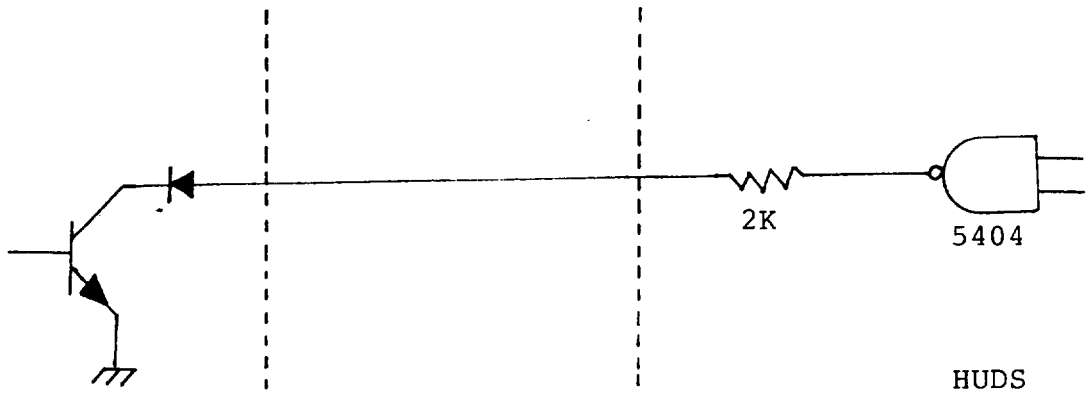
RADAR ALTIMETER

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 41

NOMENCLATURE	HIGH ALTITUDE LIMIT
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	(SEE REMARKS)
SOURCE	RADAR ALTIMETER - APN-209
SOURCE IMPEDANCE	N/A
MAX CURRENT	10 ma
MAX VOLTAGE	5.5 VDC
TRUE STATE VOLTAGE	TRANSISTOR SWITCH CLOSED (ALTITUDE BELOW LIMIT)
FALSE STATE VOLTAGE	TRANSISTOR SWITCH OPEN (ALTITUDE ABOVE LIMIT)

REMARKS



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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 42

NOMENCLATURE	FIXED FORWARD GUN MODE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE
ORIGIN	AIRCRAFT POWER
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geq$6.6K OHMS
SOURCE	PILOT'S ACTION SWITCH
SOURCE IMPEDANCE	$\leq$1 OHM
MAX CURRENT	5 ma
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	28 VDC
FALSE STATE VOLTAGE	OPEN

REMARKS

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 43

NOMENCLATURE	SPARE DISCRETE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE
ORIGIN	
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geq$6.6K OHMS
SOURCE	TBD
SOURCE IMPEDANCE	$\leq$1 OHM
MAX CURRENT	5 ma
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	+28 VDC
FALSE STATE VOLTAGE	OPEN

REMARKS

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 44

NOMENCLATURE	DIRECT/INDIRECT ROCKET MODE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE UNSHIELDED
ORIGIN	
DESTINATION	FCC
DESTINATION IMPEDANCE	10K OHMS ±20%
SOURCE	N/A
SOURCE IMPEDANCE	1 OHM
MAX CURRENT	5 ma
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	28 VDC (INDIRECT)
FALSE STATE VOLTAGE	OPEN (DIRECT)

REMARKS

Used by FCC to compute fire control reticle location for direct and indirect fire mode.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 46

NOMENCLATURE	SPARE DISCRETE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE
ORIGIN	AIRCRAFT 28 VDC
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	$\geq$6.6K OHMS
SOURCE	TBD
SOURCE IMPEDANCE	$>$1 OHM
MAX CURRENT	5 ma
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	28 VDC (DISPLAY SYMBOL)
FALSE STATE VOLTAGE	OPEN

REMARKS

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DIGITAL SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 48

NOMENCLATURE	HUD DATA
TYPE	BINARY 2's COMPLEMENT
CONDUCTOR TYPE AND GAUGE	TWISTED PAIR SHIELDED
ORIGIN	
DESTINATION	HUDS/SIGNAL PROCESSOR
DESTINATION IMPEDANCE	SEE REMARKS
SOURCE	FCC
SOURCE IMPEDANCE	≤100 OHMS
ADDRESS	N/A
NO. OF BITS/WORDS	8 BITS/1 PARITY/1 START/ 1 STOP=11 BITS
FIRST TRANSMITTED BIT	START BIT
SCALE FACTOR	N/A
LSB VALUE (RESOLUTION)	SEE TABLE I, FIGURES 2 and 3
PARAMETER RANGE	SEE TABLE I, FIGURES 2 and 3
PARAMETER ACCURACY	SEE TABLE I, FIGURES 2 and 3
TRANSMISSION RATE	9600 BAUD
ITERATION RATE	50 Hz
TRUE STATE VOLTAGE	TTL COMPATIBLE
FALSE STATE VOLTAGE	TTL COMPATIBLE

REMARKS

This signal must be compatible with Fairchild 9615, or equivalent, differential TTL receiver.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 49

NOMENCLATURE	LEFT MISSILE FIRING
TYPE	DC
CONDUCTOR TYPE AND GAUGE	TWO SINGLE-CONDUCTORS
ORIGIN	
DESTINATION	HUD
DESTINATION IMPEDANCE	N/A
SOURCE	TMS
SOURCE IMPEDANCE	$\geq$200 OHM
MAX CURRENT	50 ma
MAX VOLTAGE	N/A
TRUE STATE VOLTAGE	GROUND $\pm$1V
FALSE STATE VOLTAGE	OPEN

REMARKS

A left missile is to be fired from the aircraft. The selection of a missile will ground the left signal conductor depending on its launch side position.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 50

NOMENCLATURE	RIGHT MISSILE FIRING
TYPE	DC
CONDUCTOR TYPE AND GAUGE	TWO SINGLE-CONDUCTORS
ORIGIN	
DESTINATION	HUD
DESTINATION IMPEDANCE	N/A
SOURCE	TMS
SOURCE IMPEDANCE	$\geq$200 OHM
MAX CURRENT	50 ma
MAX VOLTAGE	N/A
TRUE STATE VOLTAGE	GROUND $\pm$1V
FALSE STATE VOLTAGE	OPEN

REMARKS

A right missile is to be fired from the aircraft. The selection of a missile will ground the right signal conductor depending on its launch side position.

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DIGITAL SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 52

NOMENCLATURE	RAW LRF DATA
TYPE	ASCII 7 BIT
CONDUCTOR TYPE AND GAUGE	TWISTED PAIR SHIELDED
ORIGIN	TOW/TSU-LRF
DESTINATION	HUDS
DESTINATION IMPEDANCE	SEE REMARKS
SOURCE	TOW/TSU-LRF
SOURCE IMPEDANCE	100 OHMS
ADDRESS	N/A
NO. OF BITS/WORDS	8 BITS/1 PARITY/1 START/ 2 STOP=12 BITS
FIRST TRANSMITTED BIT	START BIT
SCALE FACTOR	N/A
LSB VALUE (RESOLUTION)	SEE FIGURE 4
PARAMETER RANGE	SEE FIGURE 4
PARAMETER ACCURACY	N/A
TRANSMISSION RATE	9600 BAUD
ITERATION RATE	SEE FIGURE 4
TRUE STATE VOLTAGE	TTL COMPATIBLE (SEE FIGURE 2)
FALSE STATE VOLTAGE	TTL COMPATIBLE (SEE FIGURE 2)

REMARKS

This signal must be compatible with Fairchild 9615, or equivalent, differential TTL receiver. This signal will be continuously sent to the HUD. The HUD will display raw laser range data when "FCC GOOD" (discrete #20) false state voltage occurs.

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 98

NOMENCLATURE	HUD GO/NO-GO
TYPE	BIT DISCRETE
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE
ORIGIN	
DESTINATION	AIRCRAFT CAUTION PANEL
DESTINATION IMPEDANCE	$\geq$560 OHMS
SOURCE	HUDS
SOURCE IMPEDANCE	10 OHM
MAX CURRENT	60 ma
MAX VOLTAGE	28 VDC (MIL-STD-704)
TRUE STATE VOLTAGE	+28 VDC = GOOD
FALSE STATE VOLTAGE	OPEN = NO/GO

REMARKS

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DISCRETE SIGNAL INTERFACE REQUIREMENTS IDENT NUMBER 118

NOMENCLATURE	DISPLAYS BRIGHTNESS MODE
TYPE	DC
CONDUCTOR TYPE AND GAUGE	SINGLE WIRE
ORIGIN	AIRCRAFT 28 VDC
DESTINATION	HUDS/HUD
DESTINATION IMPEDANCE	$\geq$6.6K OHMS
SOURCE	PILOT STATION CONTROLS
SOURCE IMPEDANCE	$<$10 OHM
MAX CURRENT	20 ma
MAX VOLTAGE	MIL-STD-704
TRUE STATE VOLTAGE	+28 VDC (LOW LIGHT - DIM)
FALSE STATE VOLTAGE	OPEN (NORMAL VIEWING - BRIGHT)

REMARKS

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APPENDIX B

BURN IN REQUIREMENTS

Burn in is to be applied to all deliverable systems or LRU's.

Test Environments

Environmental exposure for the HUDS during burn in is Level F of MIL-STD-781B. Cycle time shall be determined by thermal survey. Chamber temperature change shall be programmed equal to or greater than 5°C per minute. Figure 6 illustrates one cycle of the environmental profile described below.

Temperature	-54°C to +71°C (-65°F to 160°F)
Temperature Cycling	Temperature cycling shall be time to stabilize at low temperature followed by time to stabilize at the high temperature, plus 2 hours.
Vibration	2.2G ± 10% peak acceleration value at 30 ± 5 Hz measured at the mounting points. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.
Equipment On-Off Cycling	Equipment off during cooling cycle and on during heating cycle.
Input Voltage	Specified voltage +5 -2%.
Input Voltage Cycling NOTE: Voltage Cycling will occur on the primary HUDS power (28VDC) only.	A. Nominal voltage for first 1/2 of "ON" time starting at the beginning of the heating period. B. Momentary off. C. 100% of nominal for next 1/3 of "ON" time. D. Momentary off. E. 90% of nominal for last 1/3 of "ON" time.

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Number of Cycles

The minimum number of cycles shall be five (5) with the last two (2) cycles failure free.

Duty Cycle Information

During equipment ON time, mode selection time shall be apportioned as follows:

Test	95% ± 5%

During operation the display and BIT Indicators shall be monitored.

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APPENDIX B

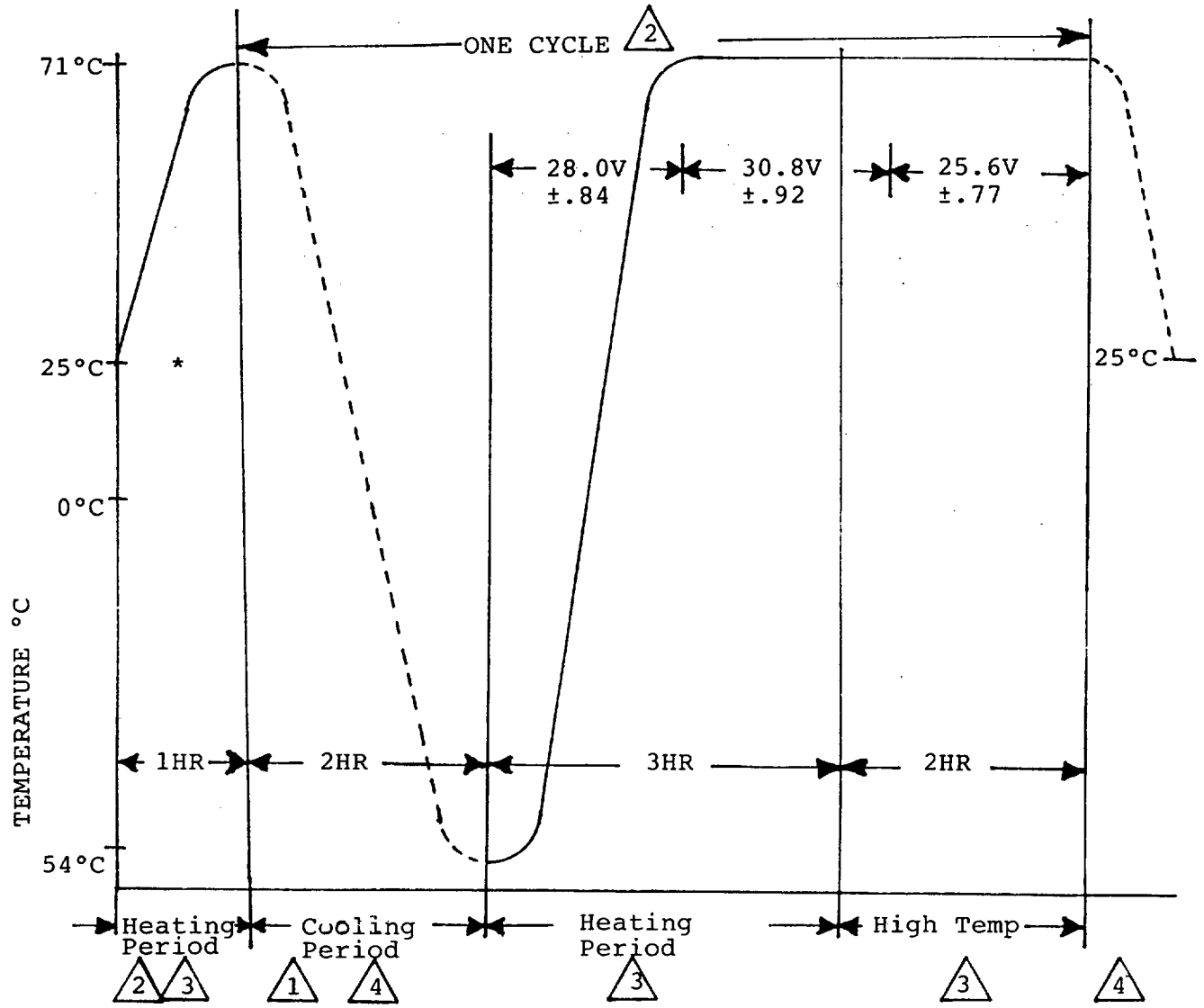


FIGURE 3-1

Typical Temperature Cycle-Electronic and Avionic Equipment

● Stabilized Temperature
O Momentary Off
---Equipment Off
——Equipment On
* To Use When Chamber is to be Dehumidified to Protect Optics.

FIGURE 6. Burn In Cycle.

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NOTES:

1. During the cooling period, the chamber temperature shall not be lower thanm -60°C.

2. One temperature cycle shall start with the equipment at high temperature and off (cooled and stabilized) at low temperature for two hours. During the cycle, the equipment shall operate in accordance with its normal duty cycle. The equipment will be on for five hours, two of which will be at high temperature. The chamber will be dehumidified when operating the HUD to protect the optics.

3. Equipment ON; input voltage (28 VDC) at 100%, 100%, and 90%, respectively, during 1/3 of the total time of heating period plus high temperature period; momentarily off between input voltage changes.

4. Equipment OFF.

5. Vibration - 2.2G ± 10% peak acceleration value at any non-resonant frequency between 20 and 60 Hertz (Hz) measured at the mounting points of the equipment. The vibration shall be at least 10 minutes during each hour of operating time.

6. Transistion rate between temperature levels shall average 5°C per minute.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-70443 (AR)	Display Unit, Sight, Helicopter Head Up Subsystem: M76

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

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b. Recommended Wording:

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